

5(a)	Electric field strength at a point is defined as the electric force per unit positive charge exerted on a small charge placed at that point.	[1]
(b) (i)	Time = $\frac{80 \times 10^{-3}}{2.50 \times 10^7} = 3.2 \times 10^{-9} \text{ s}$	[1] ans
(ii)	$\tan 30^\circ = \frac{V_y}{2.50 \times 10^7}$ $V_y = \tan 30^\circ \times 2.5 \times 10^7$ $= 1.44 \times 10^7 \text{ ms}^{-1}$	[1] sub [1] ans
(iii)	Acceleration = $\frac{V_y}{t} = \frac{(1.44 \times 10^7)}{3.2 \times 10^{-9}} = 4.5 \times 10^{15} \text{ ms}^{-2}$	[1] ans
(iv)	$a = \frac{eE}{m} = \frac{e(V/d)}{m}$ $V = \frac{adm}{e} = \frac{(4.41 \times 10^{15})(40 \times 10^{-3})(9.11 \times 10^{-31})}{1.6 \times 10^{-19}} = 1.02 \times 10^3 \text{ V}$	[1] sub [1] ans

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Suggested Solutions with Markers' Comments**

Qn	Suggested solution	Remarks
(v)	With the plate separation reduced, the electric force experienced by the charge is greater and hence the acceleration due to the force would increase . Therefore the vertical component of the velocity calculated would be greater .	
6(a)	1.6 J of energy per coulomb means the cell's emf is 1.6 V	